Self-supervised Learning
Course Introduction

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Robot Perception and Learning Lab
rpl.uni-bonn.de

What is self-supervised learning?

In this course:

Any **machine learning** technique that allows us to learn from data **without manual annotation.**

related concepts:

unsupervised learning, auto-labelling

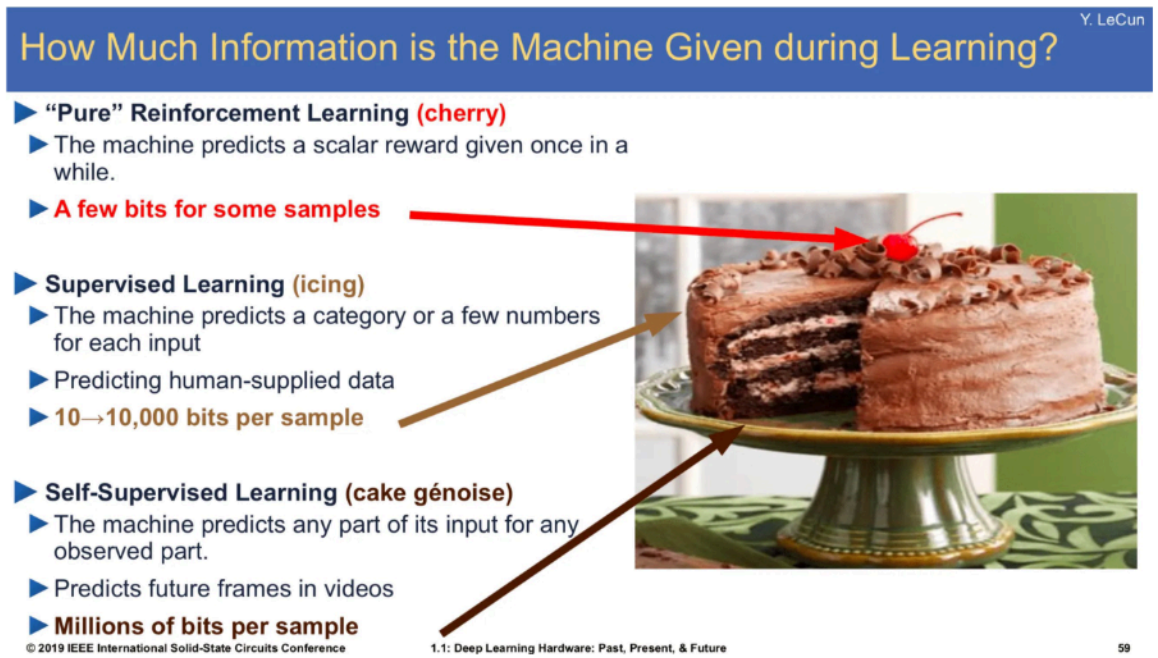
Why does it matter?

Yann LeCun‘s cake:

cherry: reinforcement learning

icing: supervised learning

filling: self-supervised learning



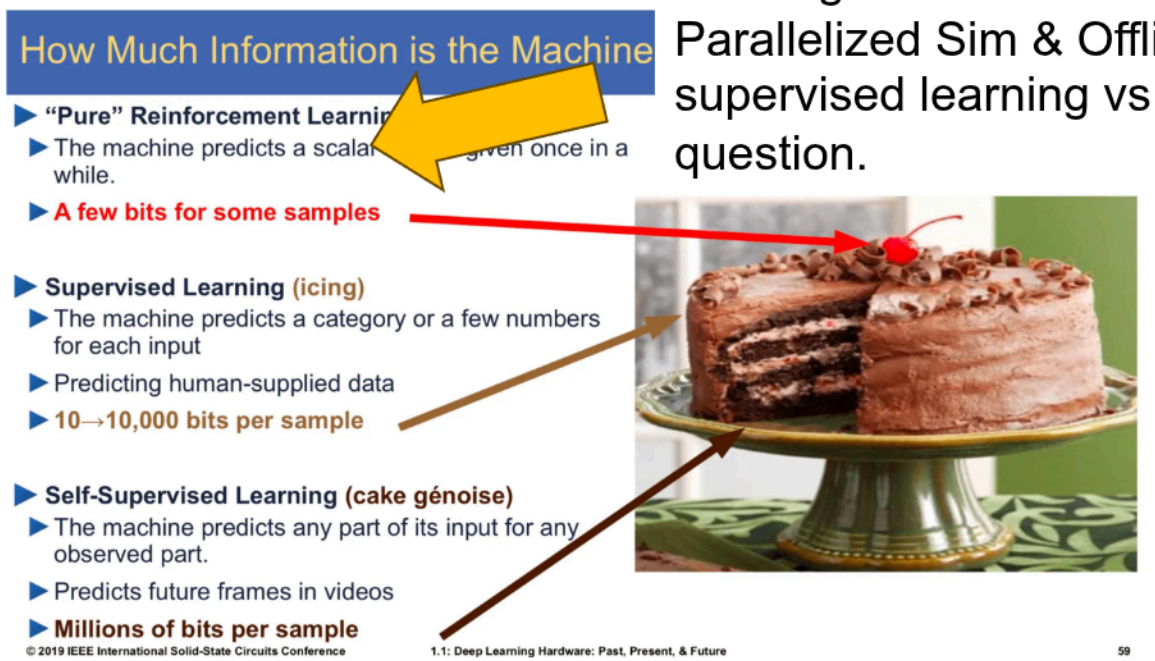
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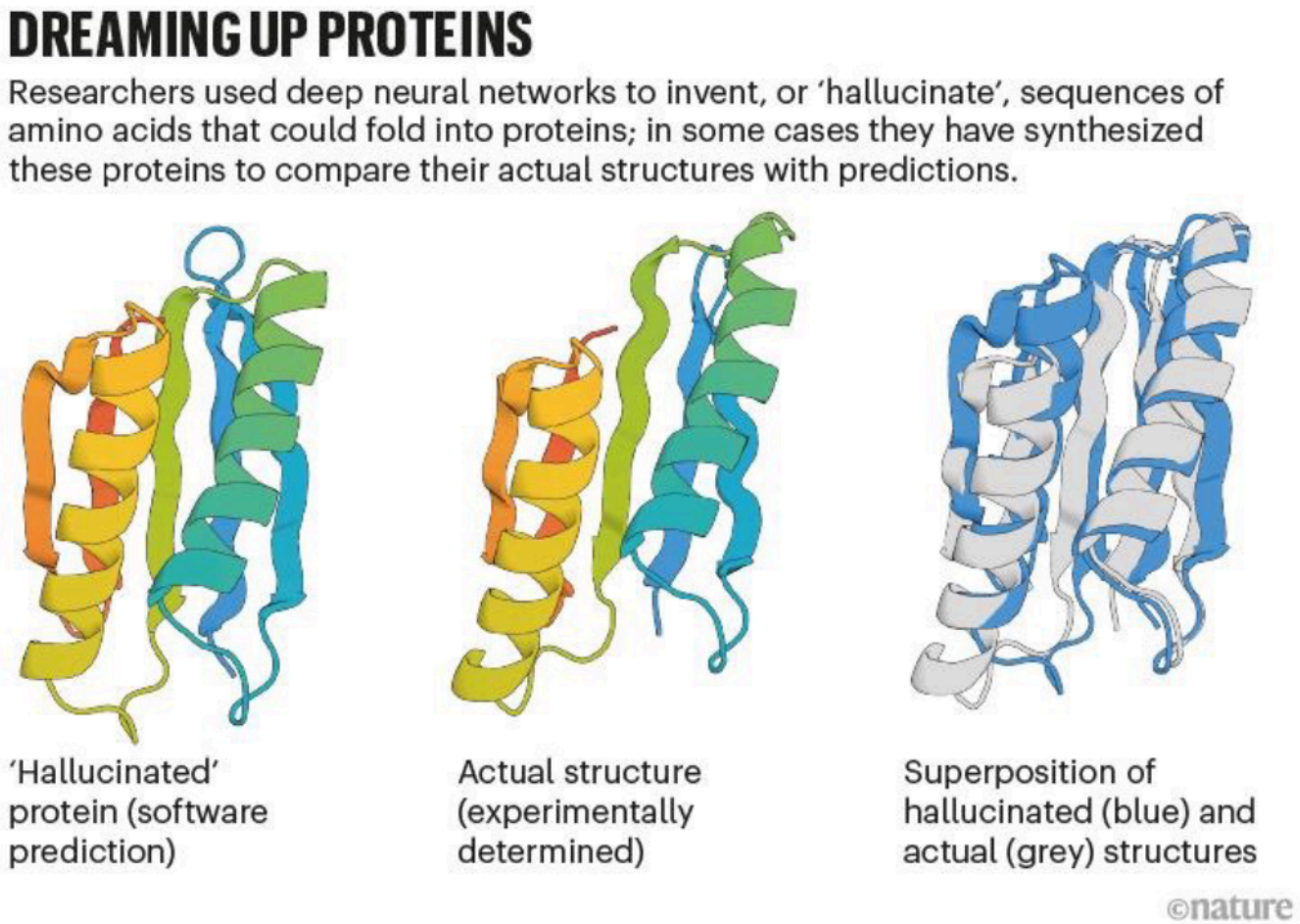
icing: supervised learning

filling: self-supervised learning



A bit unfair to reinforcement learning…
Parallelized Sim & Offline RL make supervised learning vs RL a domain question.

Example: AlphaFold



Many protein sequences are known, but 3D structures are costly to determine

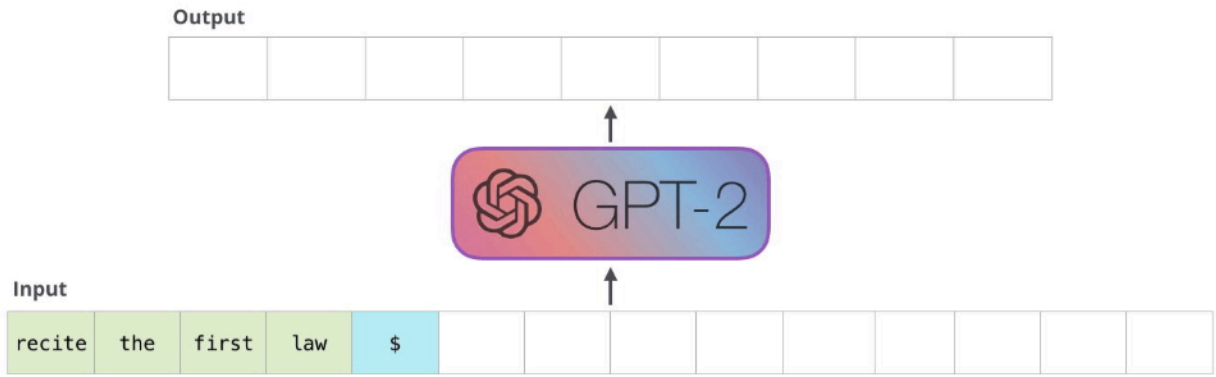
→ trains on unlabelled data with masked-element prediction and self-distillation

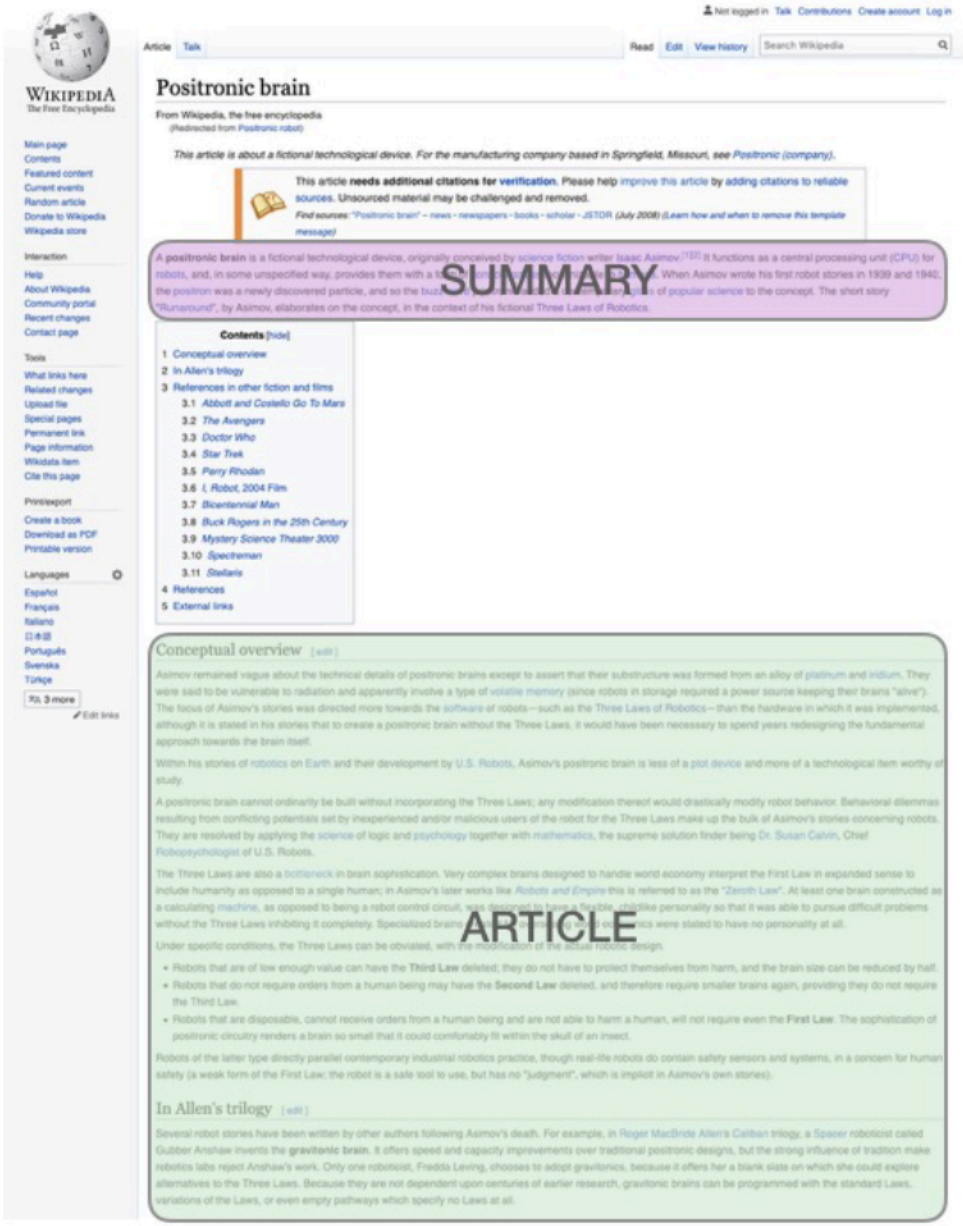
Example: Stable Diffusion



Generative models learn (complex) distributions from a set of samples, e.g. learning to generate images from samples of the distribution of „real“ pictures

Example: GPT





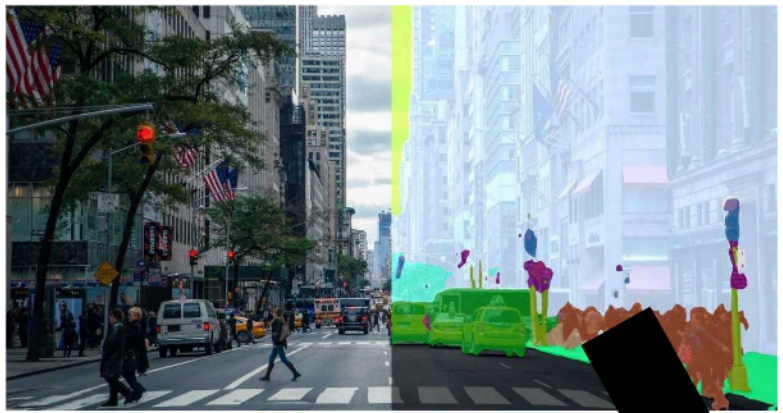
Next-Token-Prediction can be simply supervised from any text.

Also Post-Training tasks can be generated from natural structure.

Why does it matter?

Google Translate → ChatGPT

DeepLabv3



SAM3



scale

Why does it matter?



Illustration by Matt Dixon


Speed x3

Course Content

Lecture Goal:

- Students can list and explain different fundamental unsupervised, self-supervised machine learning models + different techniques of self-supervision and semi-supervision
- justify a choice of model for a given problem, evaluate if a technique is applicable
- implement it based on established software frameworks.

Methods

Nearest Neighbors, Clustering, Autoencoders, (pretrained) Neural Networks, Generative Models

Techniques

Denoising, Masking, Pre-text Tasks, Distillation, Cycleconsistency, Contrastive Learning, Pseudo-Labelling, System-level Self-Supervision

Models

GANs, Diffusion, SimCLR, DINO, [JEPA, CLIP]

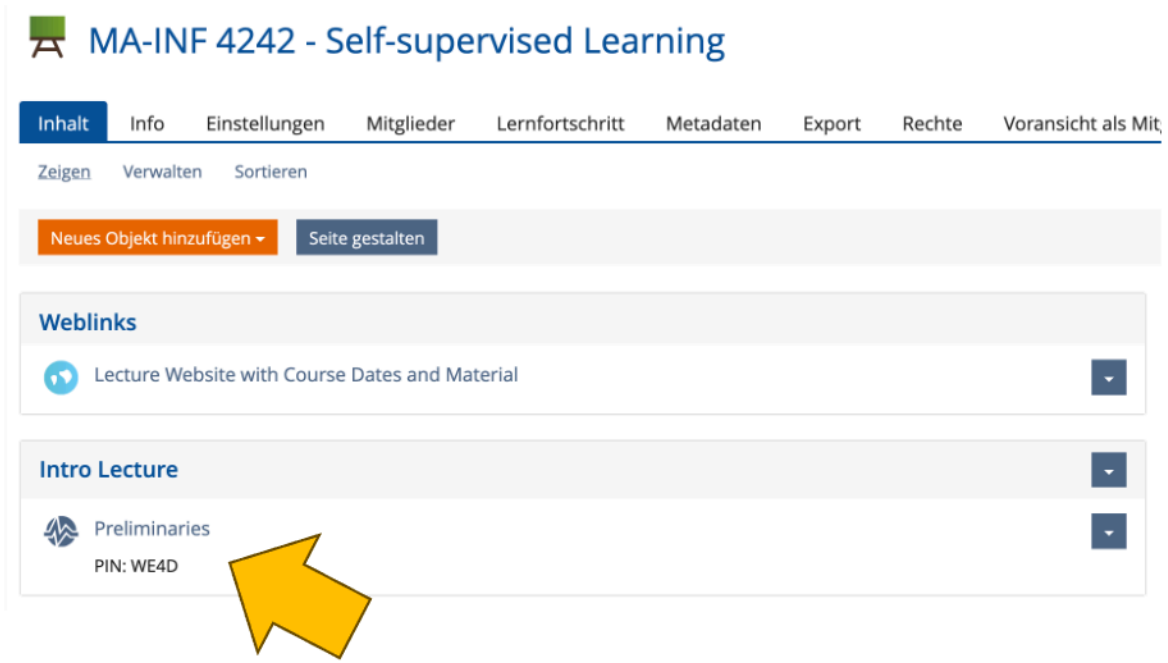
Preliminary Knowledge

This course requires a good basic understanding of deep learning and machine learning.

This is based on self-assessment.

There are no formal restrictions.

Preliminary Knowledge



ecampus.uni-bonn.de/vote/WE4D

Lecture Team

Lecture



Hermann Blum

Tutorials



Sami Azirar



Gülhan Sikaroglu



Rafay Mohiuddin

Single Point of Contact: eCampus Forum

Course Schedule

Fridays 14.15 – 15.45: Lecture, HS4

always on the following Monday, 16.15 – 18.00: Tutorial, HS1

Semester Overview · April–July 2026

April 2026

Mon	Tue	Wed	Thu	Fri
		1	2	3
6	7	8	9	10
13	14	15	16	17 L1
20	21	22	23	24 L2
27 T2	28	29	30	

May 2026

Mon	Tue	Wed	Thu	Fri
				1 Holiday
4	5	6	7	8 L3
11 T3	12	13	14	15 L4
18 T4	19	20	21	22 L5
25	26	27 Ascension break	28	29

June 2026

Mon	Tue	Wed	Thu	Fri
1 T5	2	3	4	5 No lec
8	9	10	11	12 No lec
15	16	17	18	19 L6
22 T6	23	24	25	26 L7
29 T7	30			

July 2026

Mon	Tue	Wed	Thu	Fri
		1	2	3 L8
6 T8	7	8	9	10 L9
13 T9	14	15	16	17 L10
20 T10	21	22	23	24 L11
27	28	29	30	31

Lecture · Tutorial · Public holiday · No lecture · Ascension break week

Tutorials

First Half: short test on the lecture content (not graded)
repetition and/or additional content

Second Half: practical implementation
bring your own laptop or form groups

Registration

We cannot help you with registrations from other Fakultäten or Universities. Please always ask your examination offices.

We expect you to be registered on both BASIS and eCampus.

Exam

120 min written exam

allowed material: 2 A4 sheets (both sides) of notes
relevant content: all lectures and tutorials, except for July 24

consists of
- knowledge questions
- case studies
- coding / implementation (on paper)

Resources

Course Website https://rpl.uni-bonn.de/lectures/ssl/2026/

in case of inconsistencies, the course website is more likely to be up-to-date

on the website, you find links for

- eCampus
- BASIS
- Reading Material

Don‘t Forget to Register

Until April 30: register for the lecture in BASIS

June 1-21: register for the exam

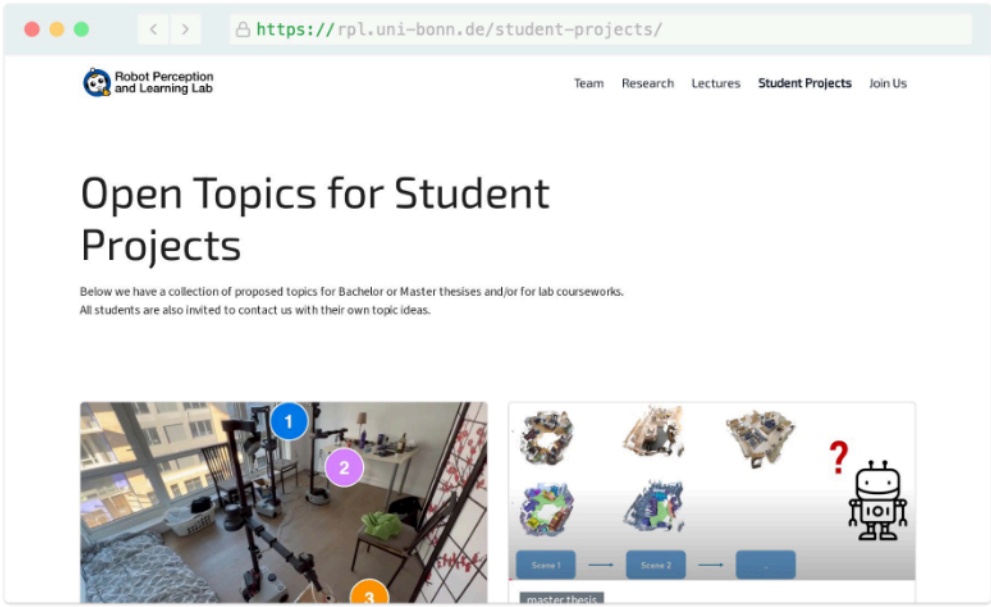
Only possible if you registered for the lecture.

You can withdraw from the exam until 7 days before.

Interested in our Research?



MA-INF-4331
Lab Perception and Learning for Robotics



Master Thesis

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Enjoy the course!

Ask questions during lecture break or on the eCampus Forum.